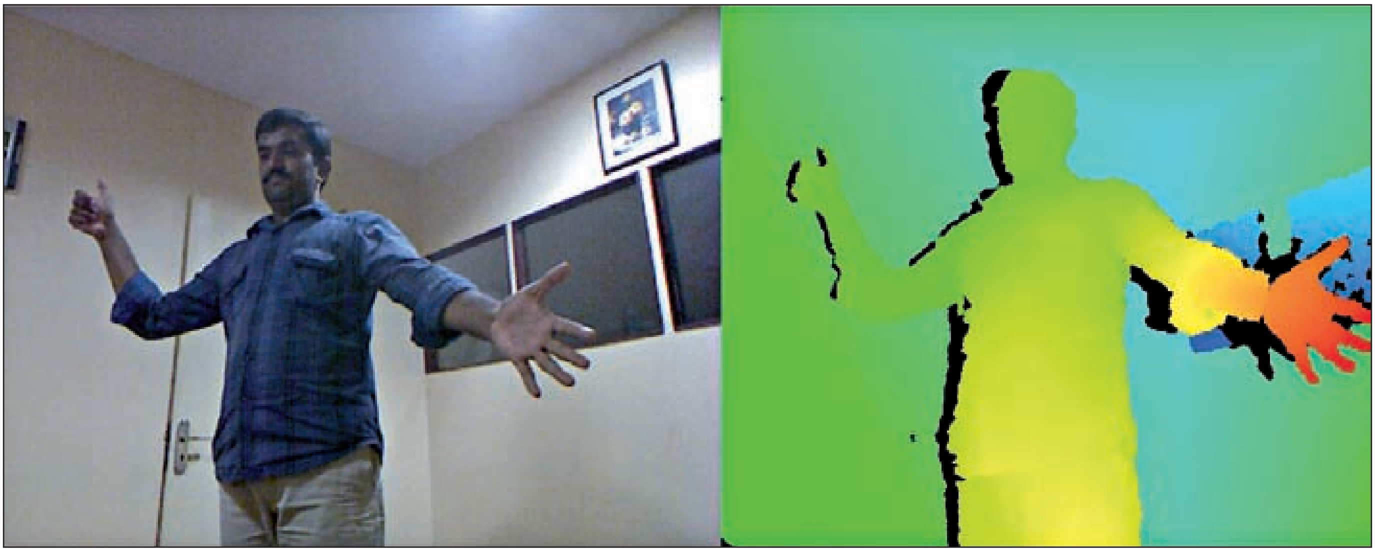

Here we see two different images of the same scene to give you an idea of how the Kinect perceives the world in front of it. On the left is the output from its VGA camera, which is close to how we see things in our surrounding. However, the image on the right shows the depth map of the same scene.

key points and then compares them in a pre-built database. The database contains the key point combinations for every human position, thereby making Kinect's feedback almost instantaneous.

IS KINECT HACK-FRIENDLY?

This may be surprising to many readers but Kinect is intentionally designed to be open, believe it or not. It isn't tough to read Kinect's output using computers or embedded devices – and there are a bunch of those out there already for users to buy. The reason behind its open nature is that most of the innovation and hardwork is inside Kinect's software system that detects skeleton position from the depth map and its database – and as things stand, it's very tough to reproduce a comparatively reliable system for skeleton detections. It works just like any USB camera with some additional information.

Interestingly, Microsoft didn't deploy any encryption or security layer on the Kinect to make it work only with Xbox. The software giant has decided to keep the device open and easily hackable – a great move for making custom gesture-enabled solution.

Moreover, Microsoft has now deemed hacking Kinect as completely legal. So honestly, what's stopping you from tinkering with it?

HOW DO WE HACK KINECT?

You need a Kinect (just the controller and not the whole Xbox) and any computer that can run a modern operating system (Windows or Mac OS X). In this case, we've used a Mac Mini just to have a portable and compact solution. If you want to do it the Windows way, you can download Kinect's SDK for Windows which is now available free for non-commercial use.

HACKING KINECT WORKSHOP

Here's how Harishankar controlled a Tata Sky set-top-box through a Kinect motion sensor attached to a Mac Mini.

1. Setting up your system: Kinect hacking doesn't require a super performance system. You can even use your old Pentium 4 machines to run the software. It's important that you decide and choose the right operating system and programming language before you begin. The decision mostly depends on whether open source modules are already available for the system you are planning to build. We chose the hardest way by trying it on our Mac Mini on which most of the code had to be ported from their Windows counterparts.

2. Reading Kinect's Output: The very first step to start hacking Kinect is reading the depth map video stream and the VGA video streams. This has been made easy by the many hackers around. There are drivers or Open-Source code available to read Kinect's data into your program. You can also play around with changing the LED indicator colours and the orientation motor which rotates the Kinect's axis.

3. Skeleton Detection: This task might be little tricky to setup, depending on the computer environment you're trying it on. You need to find the right version of the open-source software to work with your program. Writing algorithms of your own will take ages to complete unless they are crowd-sourced - ask us, we've been trying it for the past six months. You can check PrimeSense' SDKs, for instance. The thing is reliable enough to give it a try unless you're trying to build a surgical robot; hope no future doctors are disappointed. On implementing these techniques , you should get the details about a person's skeleton viz. 3D positions of each joints of the user's body.

4. Alternatives: If you chose the Windows Kinect SDK that was released recently, you can jump straight to the next step. The SDK manages to read input and detect skeletons directly. Microsoft Kinect SDK seems to be quite sophisticated for hacking, so do take a look around.

5. Gesture Recognition: Now that you have the user's positions in 3D space (x, y, and z co-ordinates), you'll have all the information regarding the user's